C08 # Attack Surface & Threats

Design Intermediate

Understanding the attack vectors specific to Claude Code

The 4 main vectors

Claude Code is an agent with access to the filesystem, Bash commands, and the internet. Its attack surface is broader than a simple text assistant.

Vector 1: Prompt injection via external content. When Claude reads an email, a GitHub issue, or an analyzed file containing malicious instructions, it may execute them. The attack exploits the confusion between data and instructions.

Vector 2: Secret exfiltration via Bash. If permissions are too broad, a malicious prompt can ask Claude to read `.env`, SSH keys, or locally stored tokens, then exfiltrate via an HTTP request.

Vector 3: Supply chain via malicious skills and plugins. A Snyk study (ToxicSkills, 2026) on 3,984 skills found 36.8% of skills with security flaws, including 534 with critical risk. The "rug pull" is the most insidious variant: a legitimate MCP that becomes malicious after being approved.

Vector 4: Privilege escalation in multi-agent pipelines. In an orchestration, a compromised agent can pass malicious instructions to subsequent agents. The outputs of one agent become the inputs of the next without intermediate validation.

Quick risk matrix

Scenario	Risk	Immediate action
Solo dev, public repos	Medium	Install an output-scanner hook
Team, sensitive codebase	High	MCP vetting + injection hooks
Enterprise, production	Critical	Least privilege, isolation and integrity verification

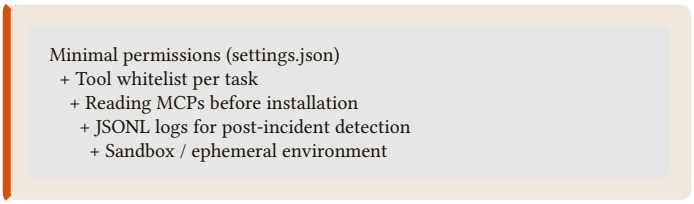
CVEs to know (selection 2025-2026)

CVE	Severity	Summary
CVE-2025-53109/53110	High	Sandbox escape filesystem MCP
CVE-2025-54135	High	RCE via prompt injection in Cursor
ADVISORY-CC-2026-001	High	Sandbox bypass, patch v2.1.34+
ADVISORY-CC-2026-002	Medium	Deny rules silently dropped past 50 subcommands, patch v2.1.90+
CVE-2026-0755	Critical (9.8)	RCE in gemini-mcp-tool (no patch)

Immediate action if you are on v2.1.33 or earlier: update to v2.1.34+ to fix the sandbox bypass.

Defense in depth

The principle is simple: each layer reduces the probability that an attack succeeds.



Basic permissions rule: grant only what the task requires. For a read/analysis task, restrict the agent with `tools: Read, Grep, Glob` or CLI `--tools "Read,Grep,Glob"`. An allow rule only pre-approves a tool; it does not remove other tools.

Audit with JSONL logs

Claude Code writes all its tool calls to the session JSONL file, `~/.claude/projects/{encoded-path}/{session-id}.jsonl`. These logs allow detecting abnormal behavior after the fact: unexpected reads of sensitive files, unplanned network calls, modification attempts outside scope.

```
# Inspect tool calls from a session
cat ~/.claude/projects/*/*.jsonl | jq '.message.content[]?.name // empty'
```